

Fig 10
Forward bias on base-collector. A low resistance should be indicated.

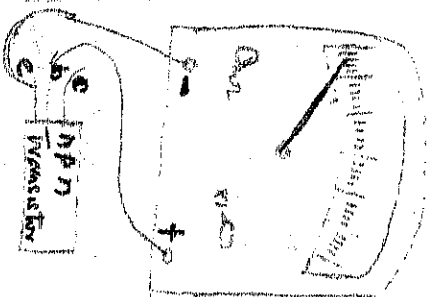


Fig 11
Reverse bias on emitter base. A high resistance should be indicated.



Fig 12
Reverse bias on collector-base. High resistance should be indicated.

HOW TO LOCATE TRANSISTOR TERMINALS AND IDENTIFY THE TYPE OF TRANSISTOR.

Having established the ohmmeter lead polarity, first identify the device leads if not known (see figures 11-12). Measure the forward and the reverse resistance between pairs of lead until you find two that measure high in both directions (over 100K Ω). These must be the collector and emitter (provided the transistor is good). The remaining one is the base.

Now measure the resistance from base to one of the other transistor leads, it should be low in one direction and high in the other.

If the low resistance occurs when the meter lead with the positive voltage is connected to the transistor base, the transistor will be n-p-n type. Of course, it will be the other way round for p-n-p.

OPEN: If either junction shows up high resistance in both directions, it is open circuit.

SHORT: Low resistance in both directions shows short circuit.

LEAKAGE: If is slightly low where is suppose to be high it shows leakage.

ELECTROLYTIC CAPACITOR.

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A good electrolytic capacitor will indicate a low resistance initially as the capacitor charges, but the resistance should rapidly increase to approach infinity.

OPEN: The resistance meter remaining infinity in the first reading after it, it will remain low ~~at the first reading~~. It is better to replace it with new one to confirm.

RESISTOR

The meter should give the correct value on meter if the zero line of the meter is well set before testing. OPEN: It will give any reading on all the ranges.

It can also increase or decrease in value by reading higher or lower than the value.

Common capacitor (non polarized).

The common (non polarized) capacitor will not read at all if it is good.

In case of short circuit, it will read very low. In it can read slightly higher in case of leakage.

~~Observation~~ It has been observed that using X1 range & an ohmmeter with a high internal battery voltage can damage a low-powered transistor.

2. Multimeter should have a relatively high resistance otherwise the loading effect of the ^{meter} could lead to incorrect conclusions.

3. Because of the shunting effect of other components connected in parallel, it is necessary to disconnect one terminal of the component being tested from the rest of the circuit. This will leave the component open on one side and the value of resistance measured will be for that component only.

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gm

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4. While making a resistance check, a zero reading indicates a short circuit, and an infinity reading indicates open circuit.



HOW TO USE DIGITAL METER TO TEST THE TRANSISTOR AND.

The hFE measurement is as follows.

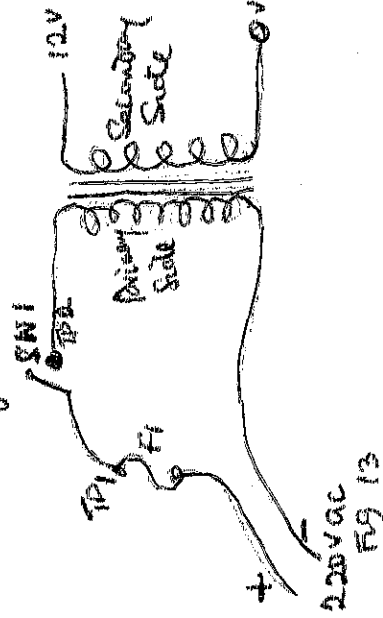
1. Set function selector switch to 200mA hFE position.
2. Insert pins of emitter, base and collector of the transistor PNP or NPN into EBC socket respectively.
3. Read the displayed value of the transistor gain (hFE) between 0 and 1000.
4. Note, the annunciator "mA" and decimal point be disregarded.

Example BC 109 transistor (PNP) gain is 46 mA.

CONCLUSION: The resistance test is very important before replacing any suspected faulty components.

USE OF MULTIMETER TO MEASURE CURRENT IN A CIRCUIT. (ANALOG OR DIGITAL)

Select the a.c. division of your multimeter, select the range higher than the voltage specified in the test point of the schematic diagram of the circuit under testing.



Connect the negative of your a.c. meter to the negative of your meter to the negative terminal of the transformer. Use the positive probe to test points 1 and 2. The meter

will indicate the amount of a.c. voltage that is passing ³² ~~through~~ the test points if the fuse on TP1 is good. ~~For~~ For the test point no 2, there will be no indication until the switch is in "on" position.

For the Secondary Side, the voltage of the meter will be on the zero while the positive ~~lead~~ will be used to test the second lead of the transformer, as the transformer is step down of 12 volts, select a range of 50-30 volts, there will be 12 volts on your meter either analog or digital. Your scale range should be ~~low~~ ^{near 250} for primary side and 30-50 volts for the secondary side.

How to Test Integrated Circuit (I.C.)

An in-circuit test is the most convenient since the power source is available and you do not have to unsolder the I.C. First measure the d-c voltage applied at the I.C. terminal to make sure that they are available and correct. If the voltages are absent or abnormal this is a good starting point for trouble shooting.

With the power sources established, the in-circuit I.C. is tested by applying the appropriate input and monitoring the output. In some cases it is not necessary to inject an input since the normal input will be supplied by the circuit ahead of the I.C.

One draw-back to testing I.C. in circuit is that the circuit before (input) and after (output) the I.C. may be defective. This could lead you to think that the I.C. was bad.

For example assume that the I.C. is an AND gate

that require two inputs to produce an output, to test such an I.C., you apply two input pulses and monitor the I.C. for an output. Now assume that the I.C. output terminal is connected to a short circuit, there will be no output indicated, even though the I.C. and the two inputs are good, off course, this will show up as an incorrect resistance measurement (if such measurements are made).

One of circuit tests for I.C.s have two obvious disadvantages, you must remove the I.C. and you must supply the required power. However if you test a suspected I.C. after removal and find that it is operating properly out of circuit, it is logical to assume that there is trouble in the circuit connected to the I.C., it is ~~better~~ ^{preferable} to correct the circuit before installation of the I.C.

TYPICAL FAULTS ON THYRISTOR CIRCUITS

1. FAULT: Gate to cathode open circuit.

RESULT AND SYMPTOMS: Thyristor off and cannot be triggered into conduction.

Measured gate signal high.

2. FAULT: Gate to cathode short circuit.

RESULT AND SYMPTOM: Thyristor off and cannot be triggered into conduction.

Measured gate signal zero.

3. FAULT: Anode to cathode short circuit

RESULT & SYMPTOM: Thyristor conducting in both forward and reverse directions.
Measured voltage drop between anode and cathode ^{is} ₁₈

FAULT: Anode to cathode open circuit.

Result & Symptom: Transistor off.

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APPACITANCE TESTER. (Practical)

To measure the electrolytic capacitor, the mode Selector Switch is in the position "Ca", the decade is out of operation, the position of the decade Switch being in materials and step the range selector Switch from the lowest value to the highest (upwards) until a deflection in the range from 10-1 appears on the blue Scale of the meter and read the measured value.

TOLERANCE MEASUREMENT.

Set the mode Selector Switch to the position "Tol" of the required range. The meter pointer moves to zero of the red Scale. When measuring L or R, connect the standard to the terminal "L" and the unknown element to the terminals "K" and read the difference on the corresponding Scale. When capacitor references are to be measured, connect the standard to the terminals "N" to maintain correct polarity of the meter deflection.

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unit switched on.

PRECAUTIONS ON TESTING COMPONENTS

SAFETY PRECAUTIONS TO BE TAKEN DURING TESTING OF EQUIPMENT.

1. Check for power supplies near the active components and in case of I.Cs directly on the appropriate pins.
2. Do not use large test probes because they can easily cause shorts.
3. Avoid the use of excessive heat when unsoldering a component and do not unsolder with the unit switched on.
4. Never remove or plug in a device without first switching off the power supply. Components can be damaged easily by the excessive surges.

COMPONENTS AND Common FAILURES.

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A Component can be said to have failed if one of its constraints is out of its specific limit. For example, if a $5k\Omega \pm 5\%$ resistor actually has a resistance value of $6k\Omega$.

There are two ways of failure ① Partial failures.
② Catastrophic failures.

PARTIAL FAILURES: This can be corrected since they do not necessarily lead to a complete loss of performance, rather to a small change. Partial failures are especially important when the Component is used in a critical circuit ^{Position.} CATASTROPHIC FAILURES. When the failure of the component is used both sudden and complete, for example, a resistor goes very high in value, or becomes open circuited or a diode develops a short circuit between anode and cathode. Such failures lead to a complete loss of performance and usually accompanied by a drastic change in device levels.

It is perhaps easy to understand failures caused by defects and overloads ~~but~~ why should a component fail in normal use?

Basically the Component is ageing because of the stresses that are acting continuously upon it. These stresses are of two kinds, operating ^{environmental} and OPERATING STRESSES is due to design conditions and the life of a component can be prolonged by operating it well within its maximum value. 21

of current, voltage and power.

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ENVIRONMENTAL STRESSSES.

These, caused by the ^{and} surrounding conditions, high temperature, high humidity, mechanical shock and vibrations, high or low pressure, corrosive chemicals, dust in the air.

Another cause of component failure is high voltage pulses or spikes generated from switches inductive loads being transmitted along the mains and appearing on internal supply wires. These spikes can easily lead to the breakdown of junctions in semi-conductor devices.

COMPONENT

COMMON TYPE OF FAULT.

Resistors — High in value and open circuit.

Variable resistors, — Open circuit, intermittent contact resulting from mechanical wear.

Capacitors

— Open circuit, Short circuit.

Inductors (including transformers)

— Open circuit, Shorted turns, Short circuit coil to frame.

Thermionic valves

— Filament open circuit, Short circuit of electrodes (ie grid to cathode) Low emissions.

Semi-conductors Devices, Diodes, transistors, FETs, SCRs

— open circuit, Short circuit at any junction. 22

VISUAL / OBSERVATION TEST METHODS.

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After the trouble is isolated to a circuit, the first step in locating the trouble is to perform a preliminary inspection using the senses, particularly those of sight, smell, hearing and touch.

For example burned or charred resistors can often be spotted by visual observation or by smell. The same holds true for oil or wax-filled components, such as capacitors, inductors and transformers when overheated the oil or wax in the case to bubble or even explode. Overheating component can be located quickly by touch. The sense of hearing can be used to listen for high voltage arcing between two wires or between wires and chassis for overheated transformers or for hum or the lack of hum - whatever the case may be.

Although it is possible to find a defective component by visual inspection, the component should not be replaced until the trouble has been investigated further. Often the real cause of trouble will destroy another component.

FAILURE SYMPTOM ANALYSIS.

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One major point on failure symptom is, do not tear into the equipment until all of the symptoms have been evaluated. One approach to failure symptom analysis is to observe the trouble symptom and then ask yourself or the operator questions regarding the symptom.

In case of total failure, the questions and answers are often obvious. For example if a television set fails to turn on, the question would be "Is there power available, is the power cord plugged in?"

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If the same television tune on but the picture is dull, the question becomes a less little obvious. Typically, you could ask "Is the brightness control turned fully on? Was the picture ever brighter than this? If the answer is yes, to both of these questions, you have definitely established failure or at least poor performance.

EVALUATING SYMPTOMS

Symptom evaluation is the process of finding a more detailed description of the trouble symptoms. The purpose of symptom evaluation is to help you understand fully what the symptoms are and what they truly indicate to point us right into the problem.

To evaluate electronic trouble symptoms, you should get as much information as possible. The mere recognition of an original trouble symptom does not in itself provide enough information to decide on the probable cause or causes of trouble.

To evaluate properly, you should manipulate the equipment controls associated with the symptom and apply your knowledge of basic electronics, supplemented with information gained from the equipment manuals.

Student

BASIC TROUBLESHOOTING REQUIREMENTS.

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Troubleshooting is a systematic, logical approach to determine and correct any fault in an equipment's operation. In its full meaning trouble shooting requires:-

1. Know in details how a given electronic equipment or system works under normal conditions. The technician should study the equipment to determine how each standard and special circuit works when operating normally. This will increase his troubleshooting efficiency.
2. Know the function of, and how to manipulate all equipment controls and adjustments. Adjusting equipment often requires re-adjustment and realignment of critical circuits. Also it would be somewhat difficult to check out a piece of equipment without knowing how to set its controls.

3. Know how to use test equipment and technical manuals or other performance data. Making voltage measurements with a multimeter set on a current would lead to a wrong observation. Technical manuals generally are the technician's best source of information containing both operational and performance data.

4. Be able to analyse logically the information of a malfunctioning equipment or a system. An analysis of the problem on a malfunctioning video

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display would help localize the trouble.

5. Be able to apply a systematic, logical procedure to locate the trouble. The time you save may be your own.

6. Be able to use proper tools to repair the trouble.

Although a pipe wrench may be used on some kinds

of micro wave plumbing, it would not fit the requirements

of a transistor radio.

7. Perform a

check-out procedure on the repaired equipment to ensure proper operation. The trouble

you repaired may have a lot of company. The

equipment may require re-adjustments because of the repair.

TECHNICAL MANUALS

A wide range of graphical material is used in the electronic industry to direct and assist with assembly wiring and alignment of equipment, most often, this material is presented as a diagram.

A diagram is a sheet of paper showing the components of an equipment and functional relationships between them by the use of suitable symbols.

Under this Standard, seven forms of diagrams are established, according to the type of elements and functional relationships involved.

In printed circuit board technique, components are inserted into holes in a phenolic board, the bottom of this board is laminated with a copper sheet that has been etched in a pattern. The remaining copper acts as the connecting wiring between the components. The entire board is passed over a wave of solder after the components are inserted, which ~~solders~~ the components to the P.C wiring and holds them in position.

Many manufacturers use printed-circuit wiring in their assembly to:

- ① Ensure uniformity of product
- ② Speed of assembly
- ③ Ease in servicing
- ④ ~~Simpler~~ locating components for servicing

Some manufacturers divide the picture of the board into grid as a road map as an aid in identifying stages and components.

The board itself is often used to give additional information for servicing.

Some manufacturers print test points, alignment frequencies, voltage readings, colour codes etc on the top or bottom of the board.

SERVICING AIDS FOR PRINTED WIRE CIRCUITS.

In service work, the most important precaution to remember when working on ~~p.c~~ PCB concern the soldering and desoldering process. Due to the nature of the laminating process, the PCB ~~must~~ must not be exposed

Step 3 Cause How to find out fault as how to procure the fault.

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To prolonged high temperature. To avoid excessive heating during servicing, you should use a small soldering iron and heat only the area required. The iron should be one of sufficient size to melt the solder quickly. The iron should always be wiped clean before desoldering. A wet sponge is best for this purpose. Use long wire pliers to hold the wire or component lead - on are soldering or desoldering.

clean the hole of solder by pushing a steel pin or round wood tooth pick through the hole. Take care not to smear the solder across other connections or leave splinters adhering to the board.

REPAIRING CRACKS IN A PRINTED CIRCUIT BOARD

A problem peculiar to PCB is hairline cracks in the board. They can be caused by abuse, such as dropping the board or they can be latent defects that appear only after repeated heating and cooling cycles. These cracks act like intermittents, making good contact while cold and opening up as the circuit heats up.

To repair the type of defects, a piece of wire should be soldered across the cracks. Just bridging the crack with solder is not sufficient since the solder will lead to separate from the board with repeated heating.

CURRENT TRACING AND VOLTAGE MEASUREMENTS.

In servicing, circuit tracing is one of the first important steps. A printed circuit board leads itself readily to circuit tracing. A lighted bulb placed 30

behind the board, on the wiring side will make the printed wiring pattern visible through the board on the component side and the circuit can be followed easily.

Voltage and resistance measurements or waveform analysis can be made by placing instrument probes on component pig tails when working on the component side of the board.

Another method is measurements at the solder points on the wiring side of the board which may be somewhat difficult because of the transparent insulating coating that is applied over the board to avoid moisture absorption in some of the PCBs.

You should use a sharp needle point probe to pierce the coating and spray the board back with transparent Krylon after the repair is completed.

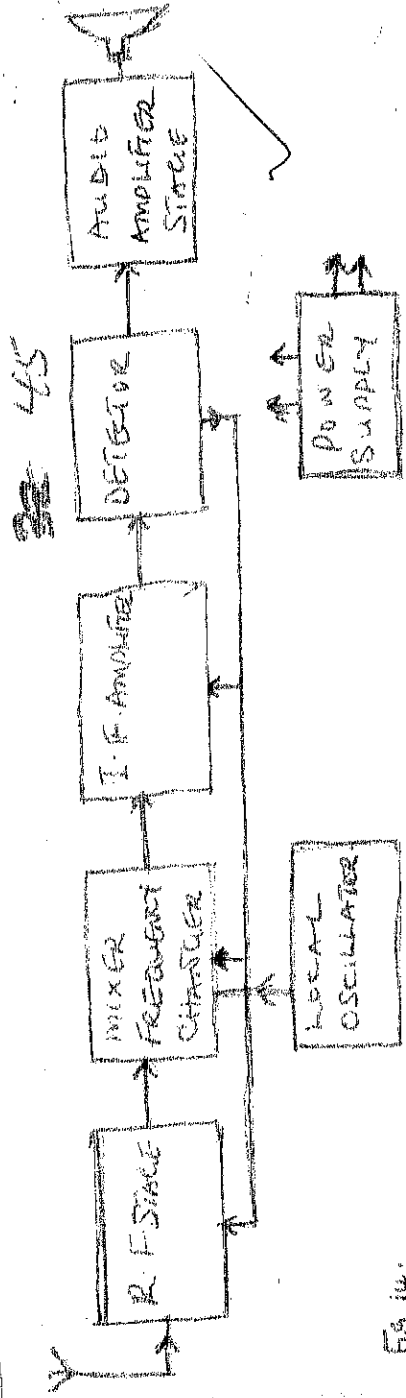


Fig 14.

BLOCK DIAGRAM OF A SUPERHETERODYNE RADIO RECEIVER.

A radio receiver consists of many parts, and to check each of these individually when a fault(s) occurs would be a very long and tedious job.

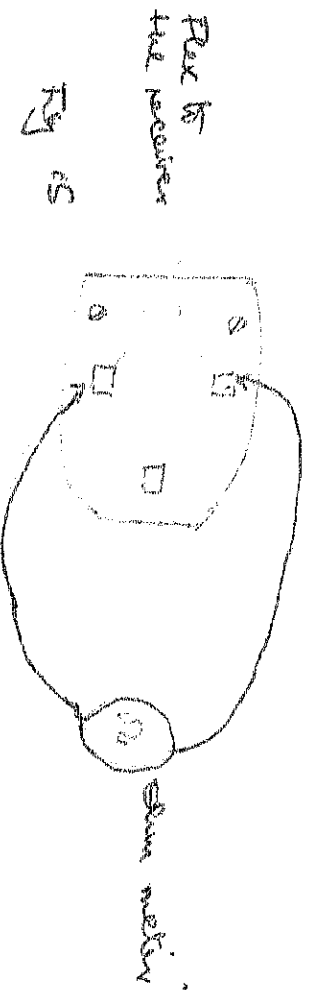
For servicing purposes, the receiver is divided into numbers of sections, each section carrying out a particular function. Consider the block diagram above. Such of these blocks has a separate function to perform, and therefore, the first step in fault finding is to find out in which of these sections the fault occurs.

The first thing to check if the receiver is completely dead is the power supply. If it is a battery powered set, then first test the battery with your meter in d.c. voltage. If the test shows that the battery is good then reconnect it and check the current that the set is drawing.

In order to protect the meter, set it first to a range of 100 mA and then, if the deflection is only small reduce it to 50 mA or 20 mA range. The current drawn by an ordinary small transistor receiver in good condition with ³² no fitted pre-set should

Certainly not exceed 20 ma and many will be between 10 and 15 ma. If a reading of 20 ma is recorded, then a fault condition is present which can quickly run down the battery. If a receiver is sent in with a completely dead battery the above fault is always present.

If the transistor receiver is supplied with power from the mains, then the checks are a little different, set the multi range meter to ohms and connect it to the mains plug as shown below.



With the receiver switch turned off, the meter will read infinity, turn on the receiver and the ohmmeter will indicate low resistance of the transformer primary. It may not be possible to give any accurate idea of what this reading should be because transformers differ so widely. The real purpose of this test is to show that as from the plug to the primary side of the transformer is in order. If the test shows less than 50 ohms then, there is a possibility that the transformer primary is faulty.

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Presuming that the above test is satisfactory, then the next step is to measure the d-c voltage that is provided by the main power components. The easiest place to measure is across either the reservoir capacitor or the Smoothing Capacitor, the value voltage obtained should be checked against the circuit diagram. If no service manual is available then look at the voltage marked on either the reservoir or Smoothing capacitors. If these are marked say ~~25~~ 25 volts, then a voltage of between 18 and 23 volts may be expected.

If the voltage is very much lower than this, then it will indicate that the receiver is drawing too much current, and alternatively if the voltage is right up to the working voltage marked on the capacitor, then it indicates that the receiver is not drawing any current at all.

If the above test is satisfactory, the next procedure is to start testing individual sections of the receiver. A signal generator will be required which has both audio and radio frequencies outputs, both of which can be adjusted to give the output desired. Select audio frequency output, set it about 1 volt and connect the leads across the volume control. A clear tone about 600 Hertz will be heard in the loudspeaker, if the audio stages are working correctly.

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BASIC STEPS IN FAULT FINDING.

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- A systematic approach to fault location should be used, the aim is to reduce the number of checks necessary to identify the fault and so reduce the time taken. The logical thought processes used in such approach can be summarized on its own right. These steps are as follows:
1. Collection of evidence
 2. Analysis of evidence
 3. Location of fault
 4. Determination and removal of the cause.
 5. Rectification of fault
 6. Checking the system.

COLLECTION OF EVIDENCE.

In collection of evidence from the user, the expression must be on its relevance to problem rather than quantity, otherwise there is a danger of missing the essential points in collecting a mass of unrelated information.

Such evidence should include the behaviour of equipment at the time and the maintenance records.

ANALYSIS OF EVIDENCE.

The evidence collected from above, should be divided into groups, in order to expose any inconsistency, which might have been presented by the user. It is here, that the detection of time to logical ~~thought~~ thought will save time.

LOCATION OF FAULTS

Here you make use of the previous steps practically. Generally, questions which may narrow the area of possible failure, are now suggested in the mind of the technician. The examination of specific sections of block can be made after the general area of fault has been found. For example, if there is no sound from the television set, the information is needed that the fault actually lies in the speaker before it is condemned.

Fault location involves, identifying the stage or area containing the fault. There may be related causes.

For example, a wire that disconnects from a terminal is an obvious reason for an open circuit and lengthy finding procedures are not necessary.

DETERMINATION AND REMOVAL OF THE CAUSE

This is an important step and one which is always omitted. An example is that of a blown main transformer in a power supply stage of the television caused by a shorted rectifier diode. The first time, the transformer may be identified and replaced. This may however be a waste of the money cause of the fault is now replaced. The later identification and removal of the primary cause may not be so easy with the possibility of finding a cause

LOCATION OF FAULTS ✓

Here you make use of the ~~previous~~ steps practically. Generally, questions which may narrow the area of possible failures, are now suggested in the area of the technician / ~~engineer~~ ^{engineer}. The examination of specific sections / blocks can be included after the general area of fault has been found. For example, if there is no sound from the television set, the information is needed that the fault actually lies in the speaker before it is condemned.

Fault location involves, identifying the stage or area contains the fault. There may be related causes. For example, a wire that disconnects from a terminal is an obvious reason for an open circuit and lengthy finding procedures are not necessary.

DETERMINATION AND REMOVAL OF THE CAUSE.

This is an important step and one which is always omitted. An example is that of burnt main transformer in a power supply stage of the television caused by a shorted rectifier diode. The first time, the transformer may be identified and replaced. This may however be in vain, if the primary cause of the fault is not replaced. The determination and removal of, the primary cause may not be so easy but the principle remains the same.

Rectification of Fault

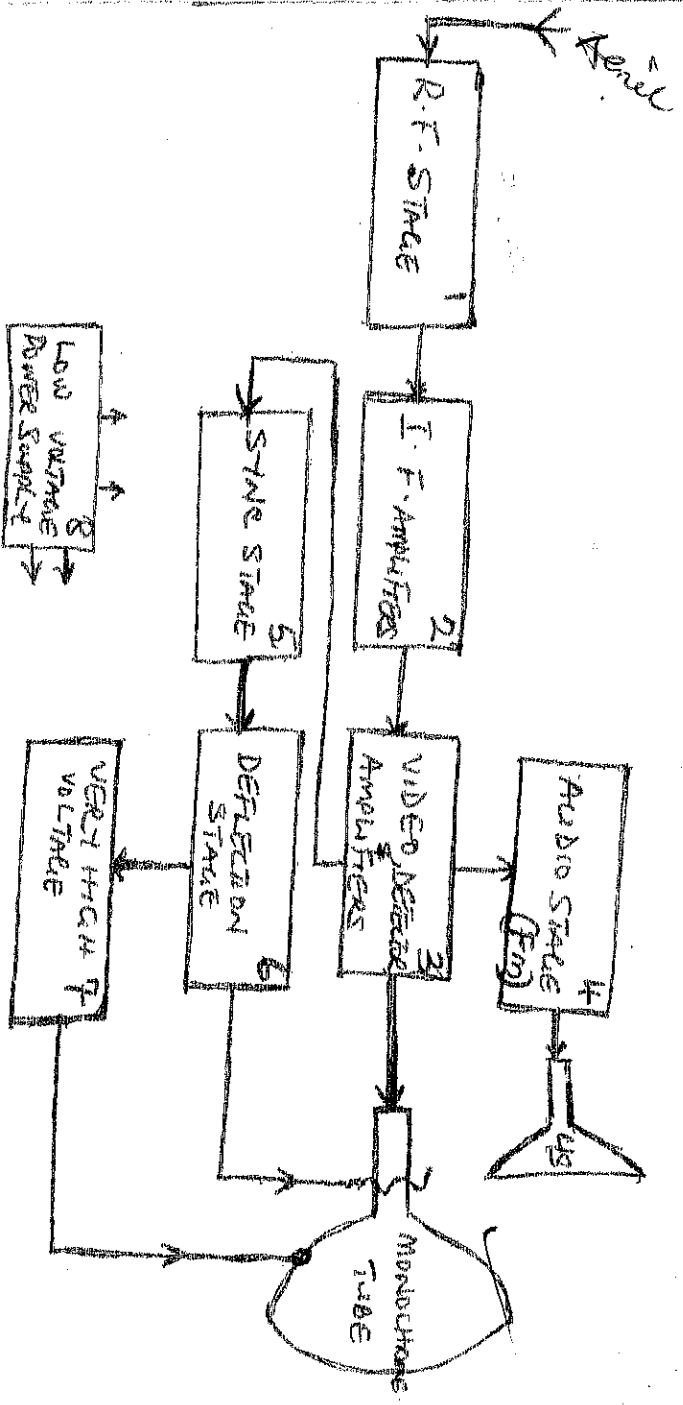
This is usually straight forward and in case of electronic small or big set, consists of anything from simple small adjustment to the replacement of some major parts.

Checking the system:

This is another step which is sometimes neglected. If a systematic approach to fault finding is not ~~used~~ adopted. There may be more than one fault in the receiver, Checking the system must always be carried out because adjustment is one step in part may well produce side effect elsewhere.

HOW TO IDENTIFY FUNCTIONAL STAGES P#4/CART ON A PRINTED CIRCUIT BOARD.

- ① The schematic diagram is your best teacher.
- ② See if there is manufacturers identification wire or colour portions to show each stage on one or both sides of your circuit board.
- ③ Know the common components in each functional stage eg volume control, speaker cone from the audio stage. Mains transformer, bridge rectifying diodes are from the power stage of the set.
④ Connective from some functional stages will serve as the input signal / power supply to some stages. eg The EHT cable to the 2nd anode of a ~~total~~ cathode-ray tube.
The R.F. Screen cable from Tuner to the I.F. stage. The speaker connections will direct you to the audio stage.



Block Diagram of a Black & White Television Receiver

Essential Components of the Television Receiver

1. **Aerial:** The starting point in the reproduction process is the aerial. For it is here that the wanted signal is collected from the air before being applied to the receiver.
2. **Tuner:** The function of a tuner is to pick out from the range of channels selected by the aerial the particular sound and vision signals required by the viewer and to amplify them to a suitable level. The second function is to lower the frequencies of the sound and vision signals to intermediate frequencies.
3. **I.F. Amplifiers:** The function of this stage is to amplify signals fed in from the tuner until they are large enough to be handled effectively in later sections.
4. **Video Section:** The video of all television receivers consist essentially of three stages, the video detector, video amplifiers and auto matrix gain control (A.G.C.) circuit.

The task of the video detector is to remove carrier, the video modulation carrying the picture information, and pass it to the video amplifier. After completion, the video signal is applied to the cathode of the picture tube where it is used to modulate the intensity of the scanning beam in such a way as to build up the desired picture on the screen. The sync circuit does its job by producing a negative volt of the appropriate value, which varies the bias conditions of the valves to which is applied. The control circuit is situated on the outside of the receiver.

SOUND SECTION The signal arrives from VHF tuner and the sound I.F. amplifiers direct. In the UHF system, the 6 mHz intercarrier from the vision detector arrives in the sound section via sound I.F. amplifiers to the VHF systems. Both the audio signals VHF and UHF are then amplified in a sound power amplification section common to both systems and are routed to the diaphragm of the loudspeaker.

SYNC SECTION The signal entering the sync section ^{VHF} is different from the video signal applied to the cathode of the picture tube. This latter signal is simply divided up so as to flow in unequal proportions to the two sections. A few rays have been separated, both line and field sync pulses are put through separate amplitude. THE FIELD SCAN SECTION The function of this section is to produce a wave form which is capable of causing the scanning beam of the picture fifty times in every second. Nearly all TV picture tubes employ the

electromagnetic methods of deflecting the electron down and across the face of the tube.

THE LINE SCAN SECTION.

The function of this is to produce a wave form which is capable of causing the scanning beam of the picture tube to move across the screen at the rate appropriate to the system.

The synchronised wave form is then passed in turn through an amplifier to a "buck it up" and through a matching transformer to step it down before being applied to the line scan coils fitted round the neck of the picture tube.

E.H.T. SECTION.

The job of this section is to produce the very high voltage needed at the final anode of the picture tube to draw the electron beam from the cathode of the tube to the screen.

The a.c. voltage induced across the auto transformer.

POWER SECTION: The picture tube and possibly all the valves in a television receiver need a heater supply

to enable them operate.
In power supply section the a.c. from mains is first rectified before being passed through a normal half

or full wave rectified. The dc output from the circuit is then smoothed (that is to say, it has ripples taken out of it) by being applied to an H.F. filter circuit.

OPERATING CONTROLS.

The adjustments on a television receiver are of three types.

1. The first group are those **Controls** that are normally accessible to and used by the viewers. These are called users or Customer Controls.

2. The second group of Controls are those the service technicians adjust when setting up the receiver in the customers home.

These ^{Service} Controls are usually located on the rear of the receiver.

3. The third group consists of the alignment adjustment for turning of the frequency response and bandwidth of the R.F., I.F., Sound I.F. and Colour circuits.

NOTE. The control ray tube neck components, Colour Service adjustments and alignment adjustments should not be touched.

OPERATING CONTROLS Command to black and white television circuit ATTEN.

FUNCTION

1. Enuff button to turn the power on

2. Volume Control Sound level

3. Brightness Adjust CAT light output

4. Contrast Adjust ratio of blue brightness

5. Channel Selector Tune in station

6. Fine tuning Exact station setting

7. Horizontal hold. Lock picture horizontally

8. Vertical hold. Lock picture vertically

9. Tone Audio frequency response

10. Automatic Sample adjustment

11. Video peaking - Sharpen picture detail

12. Colour - Adjustment of colour intensity

13. Tint Adjust to correct shade of colour

SERVICE

CORD TAP CS.

1. Contact breaker Receiver & after overload

2. Automatic gain control (AGC) Height of raster (bottom)

3. Vertical Age (Height) Height of raster (bottom)

4. Vertical linearity Squangle larger near top bottom of raster

5. Width

Power Supply

Audio Amplifier

CAT bias

Video Amplifier

Turner front end

R.F. Oscillator

Horizontal oscillator frequency

Vertical oscillator frequency

Audio Amplifier

Muting

Video Amplifier

Chroma Amplifier

3.58 mts phase

Power Supply

R.F. & IF bias

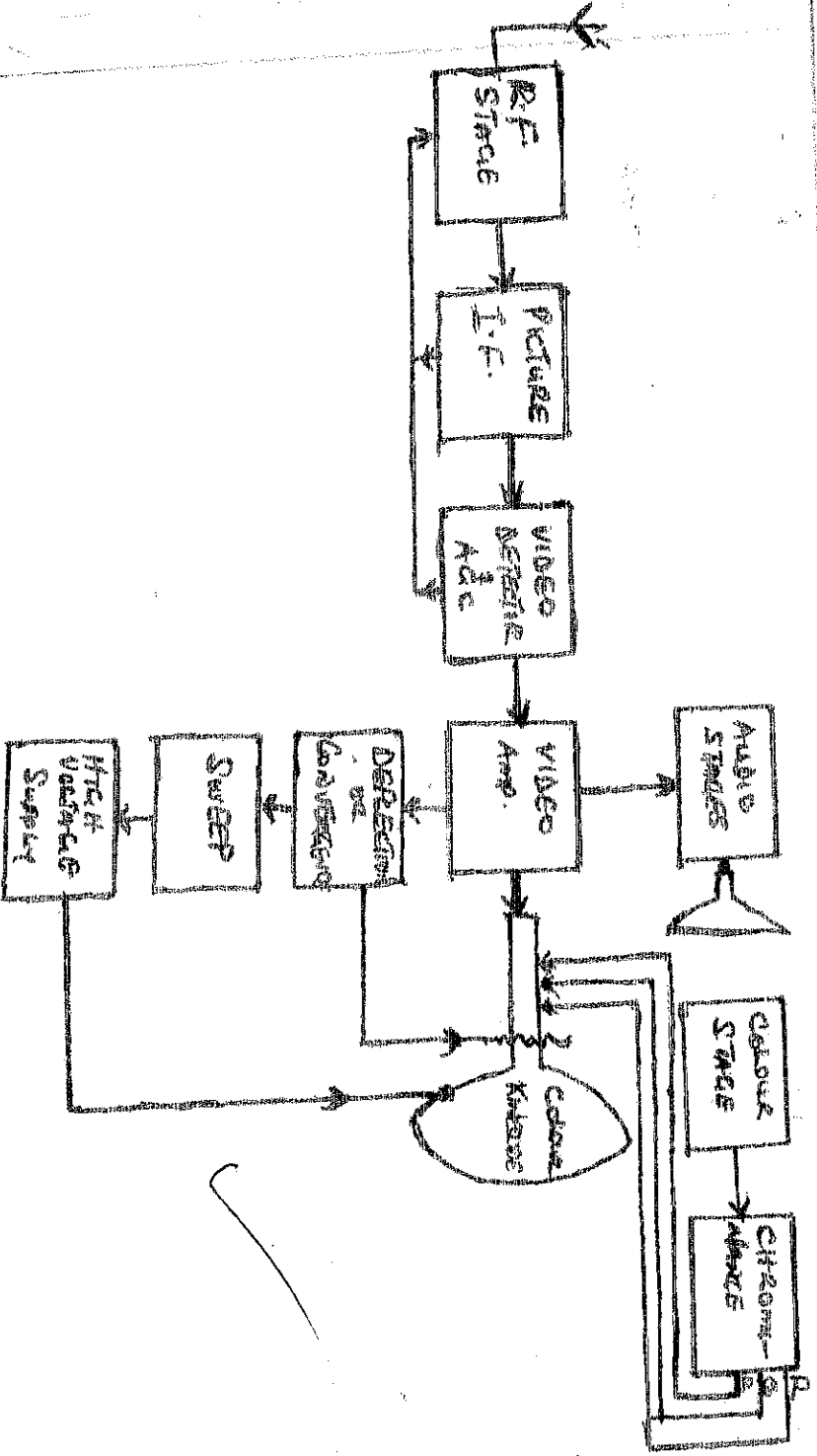
Vertical Sweep output

Vertical output stage bias

Horizontal output transformer

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6. Horizontal linearity - Square width of left and right side - Dampen waveform control (copy not efficiency).
7. Brightness range - Adjust operation of front panel control - voltage applied to brightness control.
8. A-lace inversion - Optimize sync for available signal strength - Bias or noise recover.
9. Vertical centering - Position raster vertically - Control of electron beam vertically.
10. Horizontal centering - Position raster horizontally - Control of electron beam horizontally.
11. Focus - Sharpen raster - Control to normal electron beam.
12. Pin cushion adjustment - Straighten raster line - Electric or magnetic control of raster edges (usually top & bottom).
13. Abnormal raster - out in column CRT set up -
14. ~~Service~~ Control switch may be in normal position to get a usual picture - AC and vertical sweep circuit.



Block Diagram of A Colour Television Receiver

Common Faults in Some Stages of A Colour Television

Cathode Ray Tube CRT Circuits

1. No brightness (no raster) sound okay.
 2. Low brightness
 3. Screen colour is magenta
 4. Colour in different areas of a blank and white ^{Screen}
- #### Vertical Sweep Section

1. Bright horizontal line on screen.
2. Raster or picture with vertical fringes.
3. Vertical picture more linearity
4. Loss of vertical frequency (rolling picture).

5. Vertical Key Storing.

6. Convergence errors.

HORIZONTAL SWEEP SECTION.

1. No raster

2. Dims raster

3. Insufficient width.

4. Poor horizontal linearity.

5. Horizontal foldover

6. Drive line.

7. Incorrect horizontal frequency (horizontal rolling)

8. No or poor horizontal convergence.

9. No colour.

10. Video overload due to ALC problems.

CHROMA AND COLOUR SYNC SECTION.

1. No colour

2. All bars are either blue or yellow, red or cyan.

3. No or incorrect killer actions

4. No or incorrect tint - control actions.

5. No ALC actions.

6. There are no different colours but the entire picture shifts in hue when the chroma signal is applied.